Abstract Algebra homework5

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July 29, 2026

Problem 1. Let R be a Boolean ring (every element is idempotent: $a^2 = a$ for all $a \in R$).

(a) **Claim: $x + x = 0$ for all $x \in R$.**
Since R is Boolean, $x + x$ is idempotent, so

$$(x + x)^2 = x + x.$$

Expanding the left side using distributivity (note we do *not* yet know R is commutative, so we keep the order of factors):

$$(x + x)(x + x) = x^2 + x^2 + x^2 + x^2.$$

Since every element of R is idempotent, $x^2 = x$, so

$$(x + x)^2 = x + x + x + x.$$

Combining the two displayed equations,

$$x + x + x + x = x + x.$$

Adding $-(x + x)$ to both sides gives

$$x + x = 0.$$

In particular, $x + x = 0$ means $x = -x$, i.e. every element is its own additive inverse.

(b) **Claim: R is commutative.**
Let $x, y \in R$. Since $x + y$ is idempotent,

$$(x + y)^2 = x + y.$$

Expanding,

$$(x + y)(x + y) = x^2 + xy + yx + y^2 = x + xy + yx + y,$$

using $x^2 = x$ and $y^2 = y$. Comparing with $(x + y)^2 = x + y$, we get

$$x + xy + yx + y = x + y,$$

which simplifies to

$$xy + yx = 0, \qquad \text{i.e.} \qquad xy = -yx.$$

By part (a), $-yx = yx$ (every element equals its own negative). Hence

$$xy = yx,$$

so R is commutative. ■

Problem 2. Let R be an integral domain.

(a) Fix $a \in R\backslash\{0\}$ and let $\varphi : R \to R$, $\varphi(x) = ax$. Suppose $\varphi(x_1) = \varphi(x_2)$, i.e. $ax_1 = ax_2$. Then

$$a(x_1 - x_2) = ax_1 - ax_2 = 0.$$

Since R is an integral domain and $a \neq 0$, R has no zero divisors, so $x_1 - x_2 = 0$, i.e. $x_1 = x_2$. Hence φ is injective.

(b) Suppose additionally R is finite. Let $a \in R \setminus \{0\}$; we must show a has a multiplicative inverse. By part (a), the map $\varphi(x) = ax$ is an injective function from the finite set R to itself, hence also *surjective* (an injective self-map of a finite set is a bijection). In particular $1 \in R$ is in the image of φ, so there exists $b \in R$ with

$$ab = 1.$$

Since R is commutative (integral domains are commutative rings by definition), $ba = ab = 1$ as well, so $b = a^{-1}$. As $a \neq 0$ was arbitrary, every nonzero element of R is invertible, so R is a field. ■

Problem 3. Let m be a positive integer with $m > 1$ (the natural setting for this statement; note e.g. that $\mathbb{Z}/1\mathbb{Z}$ is the zero ring and is excluded by convention from being a domain or field, and $\mathbb{Z}/0\mathbb{Z} \cong \mathbb{Z}$ is a domain but not a field, so $m = 0, 1$ are degenerate cases we set aside). We show:

$$\text{(i) } m \text{ is prime} \iff \text{(ii) } \mathbb{Z}/m\mathbb{Z} \text{ is an integral domain}$$

$$\iff \text{(iii) } \mathbb{Z}/m\mathbb{Z} \text{ is a field.}$$

(iii) ⇒ (ii): Every field is an integral domain: if F is a field and $ab = 0$ with $a \neq 0$, then multiplying by a^{-1} gives $b = a^{-1}(ab) = a^{-1} \cdot 0 = 0$. So fields have no zero divisors.

(i) ⇒ (iii): Suppose $m = p$ is prime. Let $\bar{a} = a + p\mathbb{Z} \in \mathbb{Z}/p\mathbb{Z}$ be nonzero, i.e. $p \nmid a$. Since p is prime, $\gcd(a, p) = 1$, so by Bézout's identity there exist $x, y \in \mathbb{Z}$ with

$$ax + py = 1.$$

Reducing mod p: $ax \equiv 1 \pmod{p}$, i.e. $\bar{a}\,\bar{x} = \bar{1}$ in $\mathbb{Z}/p\mathbb{Z}$. Thus every nonzero element of $\mathbb{Z}/p\mathbb{Z}$ is invertible, so $\mathbb{Z}/p\mathbb{Z}$ is a field.

(ii) ⇒ (i): We prove the contrapositive. Suppose m is not prime; since $m > 1$, m is composite, so $m = ab$ for some integers $1 < a, b < m$. Then $\bar{a}, \bar{b} \in \mathbb{Z}/m\mathbb{Z}$ are both nonzero (since $0 < a, b < m$), but

$$\bar{a}\,\bar{b} = \overline{ab} = \bar{m} = \bar{0}.$$

So $\mathbb{Z}/m\mathbb{Z}$ has zero divisors and is not an integral domain.

This closes the cycle (i)⇒(iii)⇒(ii)⇒(i), so all three are equivalent. ■

Problem 4. Let R be a nontrivial commutative ring and $I \subset R$ an ideal.

(a) **Claim:** $I = R \iff 1_R \in I$**.**
($\Rightarrow$) If $I = R$ then trivially $1_R \in R = I$.
($\Leftarrow$) Suppose $1_R \in I$. For any $r \in R$, since I is an ideal it is closed under multiplication by arbitrary ring elements, so

$$r = r \cdot 1_R \in I.$$

Hence $R \subseteq I$; combined with $I \subseteq R$, we get $I = R$.

(b) **Claim:** R **is a field** $\iff$ **the only ideals of** R **are** $\{0_R\}$ **and** R**.**
($\Rightarrow$) Suppose R is a field and I is an ideal with $I \neq \{0_R\}$. Choose $a \in I$ with $a \neq 0_R$. Since R is a field, a has an inverse $a^{-1} \in R$, and since I is an ideal,

$$a^{-1}a = 1_R \in I.$$

By part (a), $I = R$. So the only ideals are $\{0_R\}$ and R.
($\Leftarrow$) Suppose the only ideals of R are $\{0_R\}$ and R. Let $a \in R$, $a \neq 0_R$; we show a is invertible. Consider the principal ideal $(a) = aR = \{ar : r \in R\}$. Since $a = a \cdot 1_R \in (a)$ and $a \neq 0_R$, we have $(a) \neq \{0_R\}$. By hypothesis, $(a) = R$. In particular $1_R \in (a)$, so there exists $b \in R$ with $ab = 1_R$; since R is commutative, $ba = 1_R$ too, so $b = a^{-1}$. As $a \neq 0_R$ was arbitrary, every nonzero element is invertible, and R is nontrivial by hypothesis, so R is a field. ■

Problem 5. Let F be a field and R a ring, and let $\varphi : F \to R$ be a nonzero ring homomorphism.

The kernel $\ker \varphi = \{x \in F : \varphi(x) = 0_R\}$ is an ideal of F (standard fact: kernels of ring homomorphisms are ideals). By Problem 4(b), since F is a field, its only ideals are $\{0_F\}$ and F. So either $\ker \varphi = \{0_F\}$ or $\ker \varphi = F$.

If $\ker \varphi = F$, then $\varphi(x) = 0_R$ for all $x \in F$, i.e. φ is the zero map, contradicting the hypothesis that φ is nonzero. Hence

$$\ker \varphi = \{0_F\}.$$

A ring homomorphism is injective if and only if its kernel is trivial, so φ is injective. ■

Problem 6. Let $S := \mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\} \subset \mathbb{C}$. We show S is a subfield of $\mathbb{C}$.

Subring containing 1:

- $0 = 0 + 0\sqrt{2} \in S$ and $1 = 1 + 0\sqrt{2} \in S$.
- Closed under addition: $(a+b\sqrt{2})+(c+d\sqrt{2}) = (a+c)+(b+d)\sqrt{2} \in S$ since $a + c, b + d \in \mathbb{Q}$.
- Closed under negation: $-(a + b\sqrt{2}) = (-a) + (-b)\sqrt{2} \in S$.
- Closed under multiplication: using $(\sqrt{2})^2 = 2$,

$$\begin{aligned}(a + b\sqrt{2})(c + d\sqrt{2}) &= ac + ad\sqrt{2} + bc\sqrt{2} + bd \cdot 2 \\ &= (ac + 2bd) + (ad + bc)\sqrt{2} \in S,\end{aligned}$$

since $ac + 2bd, ad + bc \in \mathbb{Q}$.

So S is a subring of $\mathbb{C}$ containing 1.

Inverses of nonzero elements: Let $a+b\sqrt{2} \in S$ be nonzero, so $(a,b) \neq (0,0)$ in $\mathbb{Q}^2$.

First, $a^2-2b^2 \neq 0$: if $a^2=2b^2$ and $b \neq 0$, then $(a/b)^2=2$, i.e. $\sqrt{2} = \pm a/b \in \mathbb{Q}$, contradicting the irrationality of $\sqrt{2}$; and if $b=0$ then $a^2=0$ forces $a=0$, contradicting $(a,b) \neq (0,0)$. So $a^2-2b^2 \neq 0$.

Now compute, using the conjugate $a-b\sqrt{2}$:

$$(a+b\sqrt{2})(a-b\sqrt{2}) = a^2-2b^2 \neq 0,$$

so

$$(a+b\sqrt{2})^{-1} = \frac{a-b\sqrt{2}}{a^2-2b^2} = \frac{a}{a^2-2b^2} + \frac{-b}{a^2-2b^2}\sqrt{2}.$$

Since $a^2-2b^2 \in \mathbb{Q}\setminus\{0\}$, the coefficients $\frac{a}{a^2-2b^2}$ and $\frac{-b}{a^2-2b^2}$ are rational, so $(a+b\sqrt{2})^{-1} \in S$.

Since S is a subring of $\mathbb{C}$ containing 1 and every nonzero element of S has its inverse in S, S is a subfield of $\mathbb{C}$. ■

Problem 7. Let R be an integral domain, $X = R \times (R\setminus\{0\})$, with $(a,b) \sim (c,d) \iff ad-bc=0$.

(a) **$\sim$ is an equivalence relation.**

Reflexive: $(a,b) \sim (a,b)$ since $ab-ba=0$ (R is commutative).

Symmetric: Suppose $(a,b) \sim (c,d)$, i.e. $ad=bc$. We want $(c,d) \sim (a,b)$, i.e. $cb=da$. Since R is commutative, $cb=bc=ad=da$, as required.

Transitive: Suppose $(a,b) \sim (c,d)$ and $(c,d) \sim (e,f)$, i.e.

$$ad=bc \quad \text{and} \quad cf=de.$$

Multiply the first equation by f and the second by b:

$$adf=bcf, \qquad bcf=bde.$$

(The second uses $cf=de$, multiplied by b.) Hence $adf=bde$, i.e.

$$d(af)=d(be), \quad \text{so} \quad d(af-be)=0.$$

Since $d \in R\setminus\{0\}$ and R has no zero divisors, $af-be=0$, i.e. $af=be$, so $(a,b) \sim (e,f)$.

Hence $\sim$ is an equivalence relation. We write $\frac{a}{b}$ for the class of (a,b).

(b) **p, m descend to well-defined operations on $X/\sim$.**

First, p and m land in X: if $b,d \neq 0$, then $bd \neq 0$ since R has no zero divisors. So $p((a,b),(c,d)) = (ad+bc,bd) \in X$ and $m((a,b),(c,d)) = (ac,bd) \in X$.

Now suppose $(a,b) \sim (a',b')$ and $(c,d) \sim (c',d')$, i.e.

$$ab'=a'b, \qquad cd'=c'd. \tag{$*$}$$

We show $p((a,b),(c,d)) \sim p((a',b'),(c',d'))$ and similarly for m; it suffices (by transitivity) to check this holds when only one pair is replaced at a time, but we give the direct computation.

For m: We must show $(ac)(b'd') = (bd)(a'c')$. Indeed,

$$(ac)(b'd') = (ab')(cd') =$$

$$(a'b)(c'd) = (a'c')(bd),$$

using $(*)$ and commutativity of R. So $m((a,b),(c,d)) \sim m((a',b'),(c',d'))$.

For p: We must show

$$(ad+bc)(b'd') = (bd)(a'd' + b'c').$$

Expand the left side:

$$(ad+bc)(b'd') = ad\,b'd' + bc\,b'd' = (ab')(dd') + b(cd')b'.$$

Using $ab' = a'b$: $(ab')(dd') = (a'b)(dd') = a'd(bd')$... let us instead verify each term matches a term on the right directly.

Right side: $(bd)(a'd' + b'c') = bd\,a'd' + bd\,b'c' = a'd(bd') \cdot \frac{1}{1}$; more directly,

$$bd\,a'd' = (a'd)(bd') \text{ rearranged} = a'\,d\,b\,d'$$

$$= (a'b)(dd') = (ab')(dd') = ad\,b'd',$$

using $(*)$ $(a'b = ab')$ and commutativity, which matches the first term on the left. For the second term,

$$bd\,b'c' = b'(bd\,c') = b'\big(b(dc')\big) = b'\big(b(cd')\big)$$

[using $cd' = c'd$, i.e. $dc' = cd'$ by commutativity] $= b(cd')b' = bc\,b'd'$,

matching the second term on the left. Adding the two matched terms shows

$$ad\,b'd' + bc\,b'd' = bd\,a'd' + bd\,b'c',$$

i.e. $(ad+bc)(b'd') = (bd)(a'd'+b'c')$, as required. Hence $p((a,b),(c,d)) \sim p((a',b'),(c',d'))$.

Thus $\tilde{p}$ and $\tilde{m}$, defined by $\tilde{p}\big(\frac{a}{b},\frac{c}{d}\big) := \frac{ad+bc}{bd}$ and $\tilde{m}\big(\frac{a}{b},\frac{c}{d}\big) := \frac{ac}{bd}$, are well-defined operations on $X/\sim$.

(c) $(X/\sim,\ \tilde{p},\ \tilde{m})$ **is a field.**

Write $[a,b] := \frac{a}{b}$ for classes. We verify the field axioms; each reduces to an algebraic identity in R, checked via the criterion $(a,b) \sim (c,d) \iff ad = bc$.

Additive identity: $[0,1]$ acts as 0: $[a,b]\ \tilde{p}\ [0,1] = [a\cdot 1 + b\cdot 0,\ b\cdot 1] = [a,b]$.

Additive inverses: $[a,b]\ \tilde{p}\ [-a,b] = [ab + b(-a),\ b^2] = [0,b^2]$, and $[0,b^2] \sim [0,1]$ since $0\cdot 1 - b^2\cdot 0 = 0$. So $[-a,b]$ is the additive inverse of $[a,b]$.

Additive commutativity: $[a,b]\,\tilde{p}\,[c,d] = [ad+bc,bd] = [cb+da,db] = [c,d]\,\tilde{p}\,[a,b]$, since R is commutative.

Additive associativity: Both $\big([a,b]\,\tilde{p}\,[c,d]\big)\,\tilde{p}\,[e,f]$ and $[a,b]\,\tilde{p}\,\big([c,d]\,\tilde{p}\,[e,f]\big)$ compute (after clearing denominators using well-definedness) to

$$[\,a(df)+b(cf)+b(de)\,,\ bdf\,]$$

(up to the associativity/commutativity of R's own addition and multiplication used to expand each side), so the two agree as classes.

Multiplicative identity: $[1,1]$ acts as 1: $[a,b]\,\tilde{m}\,[1,1]=[a,b]$.

Multiplicative commutativity and associativity: Direct from commutativity/associativity of R: $[a,b]\,\tilde{m}\,[c,d]=[ac,bd]=[ca,db]=[c,d]\,\tilde{m}\,[a,b]$, and $\big([a,b]\tilde{m}[c,d]\big)\tilde{m}[e,f]=[ace,bdf]=[a,b]\tilde{m}\big([c,d]\tilde{m}[e,f]\big)$.

Distributivity:

$$[a,b]\,\tilde{m}\,\big([c,d]\,\tilde{p}\,[e,f]\big)=[a,b]\,\tilde{m}\,[cf{+}de,\ df]=\big[a(cf{+}de),\ bdf\big]=[acf{+}ade,\ bdf].$$

On the other hand,

$$\big([a,b]\tilde{m}[c,d]\big)\,\tilde{p}\,\big([a,b]\tilde{m}[e,f]\big)=[ac,bd]$$

$$\tilde{p}\,[ae,bf]=\big[(ac)(bf)+(bd)(ae),\ (bd)(bf)\big]=[acbf+bdae,\ b^2df].$$

This class equals $[acf{+}ade,bdf]$: checking the defining relation, $(acf{+}ade)\cdot b^2df$ vs. $(acbf+bdae)\cdot bdf$ agree since both sides equal $b\cdot(acf+ade)\cdot bdf$ after factoring b from $acbf+bdae=b(acf+ade)$. So the two classes are equal, giving left distributivity; right distributivity is symmetric.

Multiplicative inverses: If $[a,b]\neq[0,1]$ (i.e. $a\neq 0$, since $(a,b)\sim(0,1)\iff a\cdot 1-b\cdot 0=a=0$), then $(a,b)\in X$ with $a\neq 0$, so $[b,a]\in X/\!\sim$ is defined, and

$$[a,b]\,\tilde{m}\,[b,a]=[ab,ba]=[ab,ab]\sim[1,1],$$

since $(ab)(1)-(ab)(1)=0$. So $[b,a]$ is the multiplicative inverse of $[a,b]$.

Finally, $[0,1]\neq[1,1]$ (else $0\cdot 1-1\cdot 1=-1=0$ in R, forcing R trivial, excluded since R is an integral domain with $1\neq 0$). So $X/\!\sim$ is a nontrivial commutative ring in which every nonzero element is invertible: a field. ■

(d) **An injective ring homomorphism $i:R\to\mathrm{Frac}(R)$.**

Define $i(r):=[r,1]=\frac{r}{1}$.

Ring homomorphism:

$$i(r{+}s)=[r{+}s,1],\qquad i(r)\,\tilde{p}\,i(s)=[r,1]\,\tilde{p}\,[s,1]=[r{\cdot}1{+}1{\cdot}s,\ 1{\cdot}1]=[r{+}s,1],$$

so $i(r+s)=i(r)\,\tilde{p}\,i(s)$.

$$i(rs)=[rs,1],\qquad i(r)\,\tilde{m}\,i(s)=[r,1]\,\tilde{m}\,[s,1]=[rs,1],$$

so $i(rs)=i(r)\,\tilde{m}\,i(s)$. Also $i(1)=[1,1]$, the multiplicative identity of $\mathrm{Frac}(R)$.

Injective: If $i(r)=i(s)$, then $[r,1]=[s,1]$, i.e. $r\cdot 1-1\cdot s=0$, so $r=s$. ■

(e) **Universal property.** Let F be a field and $f : R \to F$ an injective ring homomorphism.

Existence. Define $g : \mathrm{Frac}(R) \to F$ by

$$g\Big(\frac{a}{b}\Big) := f(a)f(b)^{-1}.$$

This makes sense: since $b \neq 0$ and f is injective, $f(b) \neq f(0) = 0_F$, so $f(b)^{-1}$ exists in F.

Well-defined: if $(a, b) \sim (c, d)$, i.e. $ad = bc$, apply f (a ring homomorphism):

$$f(a)f(d) = f(b)f(c).$$

Since $f(b), f(d) \neq 0$ are invertible in the field F, multiply both sides by $f(b)^{-1}f(d)^{-1}$:

$$f(a)f(b)^{-1} = f(c)f(d)^{-1},$$

so g is well-defined on classes.

Ring homomorphism: One checks directly

$$g\Big(\frac{a}{b} + \frac{c}{d}\Big) = g\left(\frac{ad + bc}{bd}\right) = f(ad + bc)f(bd)^{-1}$$

$$= \big(f(a)f(d) + f(b)f(c)\big)f(b)^{-1}f(d)^{-1} = f(a)f(b)^{-1} + f(c)f(d)^{-1},$$

which equals $g(a/b)+g(c/d)$; similarly $g\big((a/b)(c/d)\big) = f(ac)f(bd)^{-1} = f(a)f(b)^{-1}f(c)f(d)^{-1} = g(a/b)g(c/d)$, and $g(1/1) = f(1)f(1)^{-1} = 1_F$.

$f = g \circ i$: For $r \in R$, $g(i(r)) = g(r/1) = f(r)f(1)^{-1} = f(r) \cdot 1_F = f(r)$.

Uniqueness. Suppose $g' : \mathrm{Frac}(R) \to F$ is another ring homomorphism with $g' \circ i = f$. First note $[1, b] \,\tilde{m}\, [b, 1] = [1, 1]$, so $[1, b] = i(b)^{-1}$ in $\mathrm{Frac}(R)$ (this makes sense since $i(b) = [b, 1]$ is invertible in the field $\mathrm{Frac}(R)$, its inverse being $[1, b]$). Since g' is a ring homomorphism, it sends invertible elements to their images' inverses (when those images are nonzero, which holds since $g' \circ i = f$ is injective on $R \setminus \{0\}$, as f is injective and F has no zero divisors), so

$$g'([1, b]) = g'(i(b))^{-1} = f(b)^{-1}.$$

Then for any $a/b \in \mathrm{Frac}(R)$,

$$g'\Big(\frac{a}{b}\Big) = g'\big([a, 1] \,\tilde{m}\, [1, b]\big) = g'(i(a))\, g'([1, b]) = f(a)f(b)^{-1} = g\Big(\frac{a}{b}\Big).$$

Hence $g' = g$, proving uniqueness. ■

(f) **Suppose R is itself a field.** We show there is a unique ring isomorphism $\varphi : \mathrm{Frac}(R) \to R$ with $\varphi \circ i = \mathrm{id}_R$.

Apply part (e) with $F = R$ and $f = \mathrm{id}_R$ (which is trivially an injective ring homomorphism $R \to R$). This yields a unique ring homomorphism $\varphi : \mathrm{Frac}(R) \to R$ with $\varphi \circ i = \mathrm{id}_R$, explicitly

$$\varphi\Big(\frac{a}{b}\Big) = a \cdot b^{-1}$$

(the inverse taken in the field R, valid since $b \neq 0$).

φ is injective: φ is a ring homomorphism out of the field $\mathrm{Frac}(R)$ (part (c)); it is nonzero since $\varphi(i(1)) = \mathrm{id}_R(1) = 1 \neq 0$ (as R is nontrivial). By Problem 5, nonzero homomorphisms out of a field are injective, so φ is injective.

φ is surjective: Given $r \in R$, we have $\varphi(i(r)) = \mathrm{id}_R(r) = r$, so r is in the image of φ. Hence φ is surjective.

Therefore φ is a bijective ring homomorphism, i.e. a ring isomorphism, and it is the unique one satisfying $\varphi \circ i = \mathrm{id}_R$ by the uniqueness clause of the universal property. ■

Remark. Applying part (f) to $R = \mathbb{Z}$ shows that $\mathbb{Q} := \mathrm{Frac}(\mathbb{Z})$ recovers the familiar rational numbers, consistent with the definition given at the end of the problem set.